JAMES RUSE AGRICULTURAL HIGH SCHOOL

2024

YEAR 10 SCIENCE TEST

YEARLY PAPER

Total: 80 Marks

Reading time – 5 minutes

Working time – 90 minutes

Answer questions in the spaces provided.

General Instructions

- NESA Board-approved calculators may be used.
- Write using a black pen.
- Draw diagrams using a pencil.
- Data Sheet and periodic table provided.

SECTION A (20 Marks)

Use the multiple-choice answer sheet on Page 11 for Questions 1–20.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

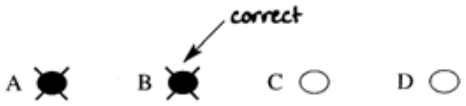
Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

A ○ B ● C ○ D ○

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ● B ⊗ C ○ D ○

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word ~~correct~~ and drawing an arrow as follows.



1. Which evidence best supports the theory that the universe was created by an explosion called the Big Bang?

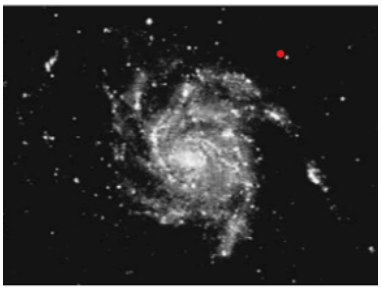
   A. Impact craters found on Earth.

   B. Cosmic background radiation.

   C. The different compositions of terrestrial and gaseous planets.

   D. The blue shift of light from distant galaxies.

2. The photograph below shows the universe as seen through a telescope.



This feature is best identified as

   A. A galaxy

   B. A comet

   C. An asteroid

   D. A star

3. Which process generates the energy that is ultimately released by stars?

A. Nuclear fusion

B. Convection currents

C. Thermal conduction

D. Radioactive decay

4 What celestial phenomenon most affects radio communication and other electrical atmospheric changes for us on Earth?

A. Solar eclipse

B. Solar flares

C. Lunar eclipses

D. Meteorites entering the atmosphere

5. What causes clouds of dust and gas to form a protostar?

A. Magnetism

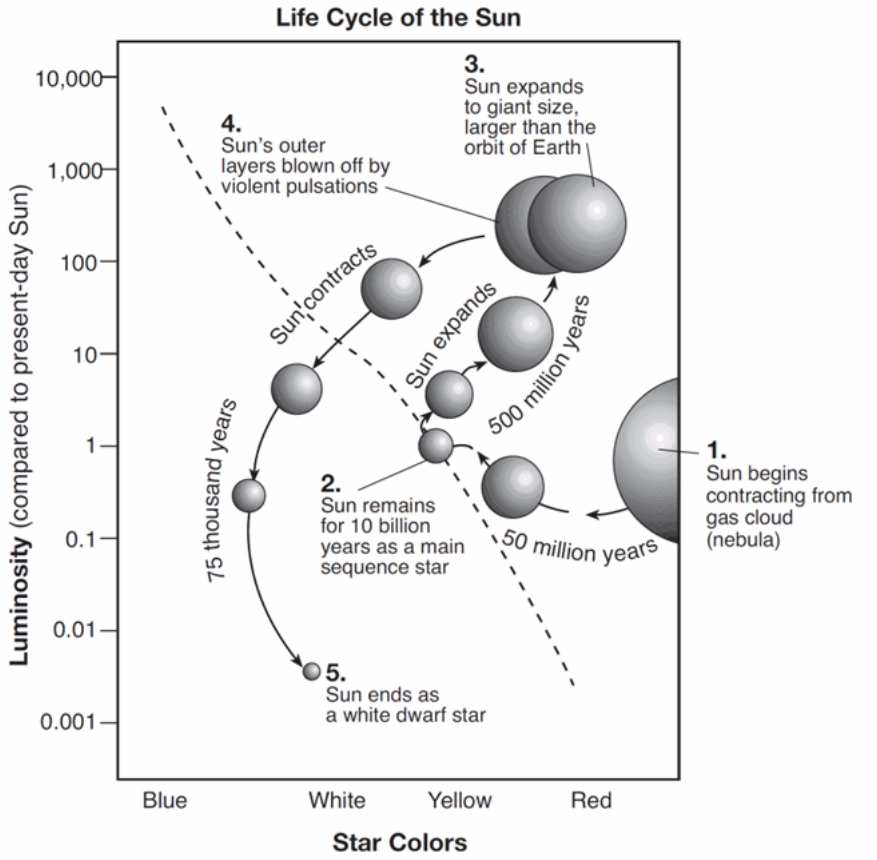
B. Gravitational attraction

C. Expansion of matter

D. Cosmic background radiation

6. Base your answer to the following question on the diagram below.

The diagram represents the inferred changes to the luminosity and colour of the Sun throughout its life cycle. The diagonal dashed line represents the main sequence stars. The numbers 1 through 5 represent stages in the life cycle of the Sun.



The Sun is inferred to be the most luminous when it is classified as a

A. White dwarf star

B. Gas cloud (nebula)

C. Main sequence star

D. Giant star

7. A student is reacting hydrochloric acid with zinc granules in a beaker. Which of the following would increase the rate of this reaction?

A. Decreasing the temperature.

B. Adding water to the beaker.

C. Using a test tube instead of a beaker.

D. Cutting the zinc into smaller pieces.

8. Which of the following best defines corrosion?

A. The process by which a metal reacts to form a metal oxide.

B. The gradual breakdown of a metal due to chemical reactions with the environment.

C. The process of metals is becoming more conductive over time due to exposure to air.

D. The formation of rust on iron after exposure to salt and water.

9. Two islands are drifting apart due to geological changes. When the islands were close together, similar populations of an insect were found on each island.

As the islands drift further apart, which process must happen for speciation to occur?

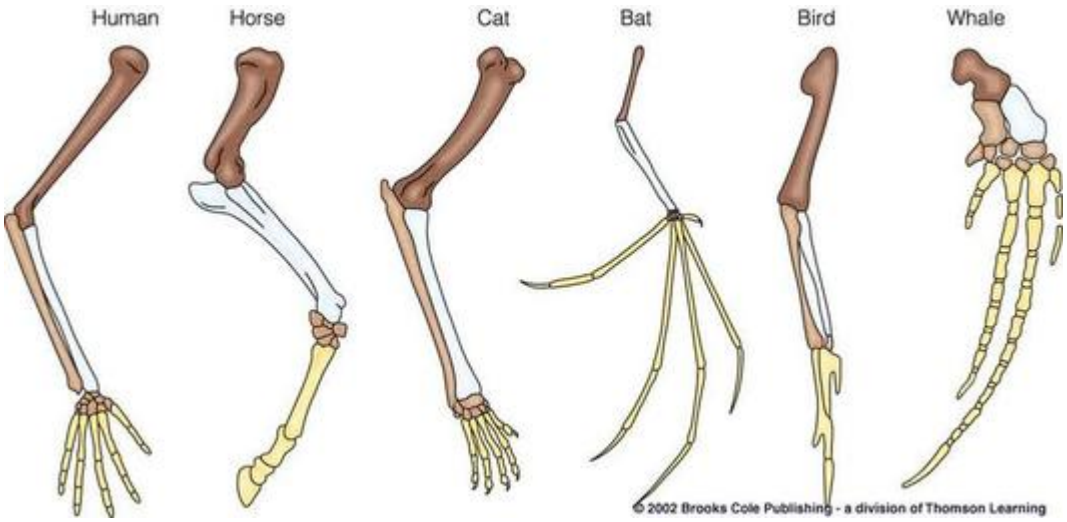
A. Different predators evolve.

B. Reproductive isolation.

C. Climate change.

D. Continental drift.

10. Consider the following skeletal structures:



Which type of evolution does this illustrate?

A. Codominant

B. Dominant

C. Convergent

D. Divergent

11. For natural selection to take place, many more offspring must be born than survive to maturity.

What else is required?

A. Inheritable characteristics that suit the environment.

B. A decrease in the number of predators.

C. Characteristics of the ecosystem change.

D. A mass extinction event occurs.

12. There is much more phenotypic variation among domesticated dog breeds than in the closely-related non-domesticated coyote.



What accounts for this observation?

A. Higher rates of mutation during domestication.

B. Random mating during domestication.

C. Domesticated dogs would have a low survival rate in the wild.

D. Domesticated dogs exhibit a high rate of growth.

13. Which of the following lists layers of the atmosphere correctly for decreasing altitude?

A. Troposphere, stratosphere, exosphere

B. Thermosphere, stratosphere, mesosphere

C. Exosphere, troposphere, thermosphere

D. Thermosphere, mesosphere, stratosphere.

14. La Niña is associated with which of the following conditions?

A. Warmer east Pacific temperatures and low trade winds.

B. Warmer east Pacific temperatures and high trade winds.

C. Cooler east Pacific temperatures and low trade winds

D. Cooler east Pacific temperatures and high trade winds

15. Which of the following can result from reduced ozone levels in the stratosphere?

A. Increase rate of droughts.

B. Increased rate of genetic mutation

C. Rising sea levels

D. More frequent tsunamis

16. Which of the following processes directly decreases the amount of phosphorus in the biosphere?

A. Mining

B. Erosion

C. Weathering

D. Excretion

17. What is a fertilised egg cell known as?

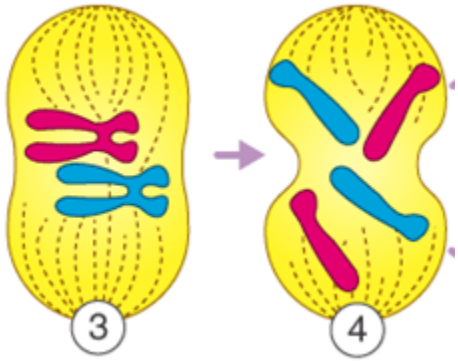
A. Gamete

B. Germ cell

C. Zygote

D. Ovum

18. The following image shows 2 stages of mitosis.



What is the purpose of the spindle fibres?

A. To provide support for the chromatids.

B. To duplicate the chromatid.

C. To pull the chromatids apart.

D. To attach the sister chromatids together.

19. Which species is being reduced in the following reaction?

$4\ Al + 3\ Fe_2O_3 \rightarrow 2\ Al_2O_3 + 6\ Fe$

A. Aluminium

B. Iron (III) oxide

C. Aluminium oxide

D. Iron

20. Which of the following is an example of an endothermic reaction?

A. Photosynthesis

B. Rusting of iron

C. Combustion of fuel

D. Respiration

ANSWER BOOKLET

Name:
Class: **10A 10C 10K 10L 10T**

Section A Answer grid for multiple choice questions 1 – 20.

1.	A O	B O	C O	D O
2.	A O	B O	C O	D O
3.	A O	B O	C O	D O
4.	A O	B O	C O	D O
5.	A O	B O	C O	D O
6.	A O	B O	C O	D O
7.	A O	B O	C O	D O
8.	A O	B O	C O	D O
9.	A O	B O	C O	D O
10	A O	B O	C O	D O
11.	A O	B O	C O	D O
12.	A O	B O	C O	D O
13.	A O	B O	C O	D O
14.	A O	B O	C O	D O
15.	A O	B O	C O	D O
16.	A O	B O	C O	D O
17.	A O	B O	C O	D O
18.	A O	B O	C O	D O
19.	A O	B O	C O	D O
20.	A O	B O	C O	D O

SECTION B (60 marks)

Answer the questions in the spaces provided.

These spaces guide the expected length of response.

Show all relevant work in questions involving calculations.

Extra writing space is provided at the back of this booklet.

If you use this space, clearly indicate which question you are answering.

Question 21 (3 marks)

Describe the role of microorganisms in the nitrogen cycle. **3**

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Question 22 (2 marks)

Outline how temperature and salinity drive the formation of deep ocean currents. **2**

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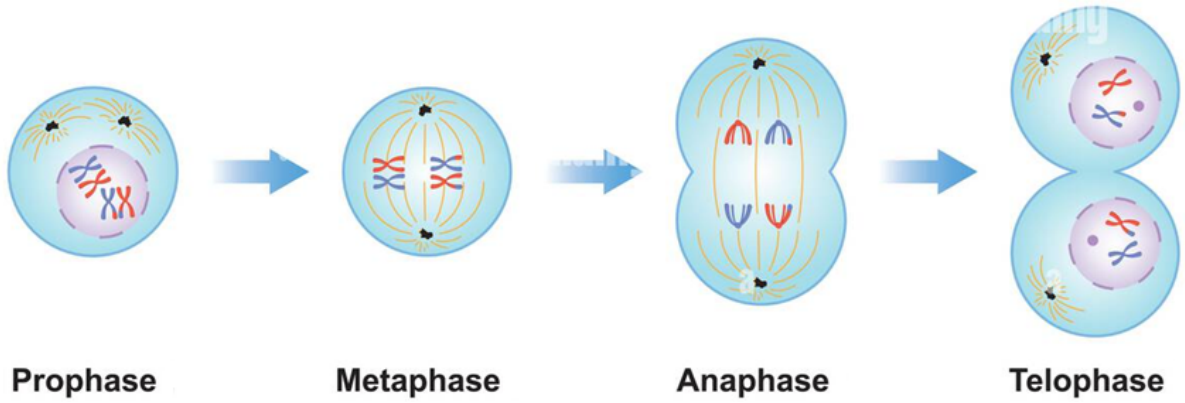
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Question 23 (5 marks)

The following diagram shows some of the stages of cell division.



(a) Identify the stage in which the chromosomes line up in the centre of the cell. **1**

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(b) Justify why this must be meiosis and not mitosis. **1**

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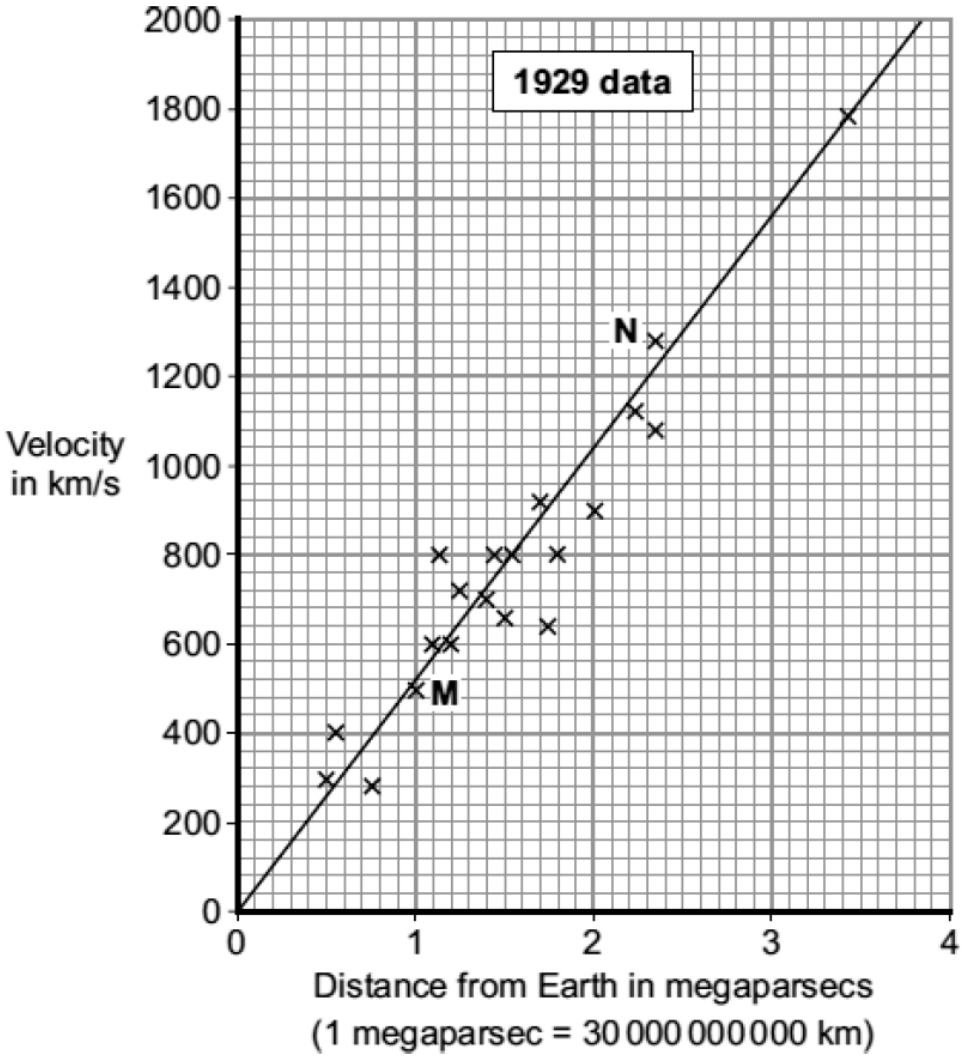
(c) Complete the following table. **3**

	Mitosis	**Meiosis**
Type of cells produced		
Diploid/haploid		
Purpose in humans		

Question 24 (6 marks)

By measuring the *red-shift*, Hubble was able to calculate the speed at which the galaxies are moving away from the Earth. He was also able to calculate the distance of these galaxies from the Earth.

The graph shows the relationship between the data collected by Hubble.



Question 24 continues on Page 15

(a) The gradient of the line drawn on the graph gives a number known as the Hubble constant. The Hubble constant can be used to estimate when the universe began.

Use the graph to calculate the value of the Hubble constant.

Show clearly how you obtained your answer.

Hubble constant = _____km /s per megaparsec **2**

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(b) The Andromeda galaxy is not moving away from the Earth. It is actually moving towards the Earth. This means that the light from Andromeda shows a blue-shift.

How do the wavelength and frequency of the light from Andromeda seem to have changed when viewed from the Earth? **2**

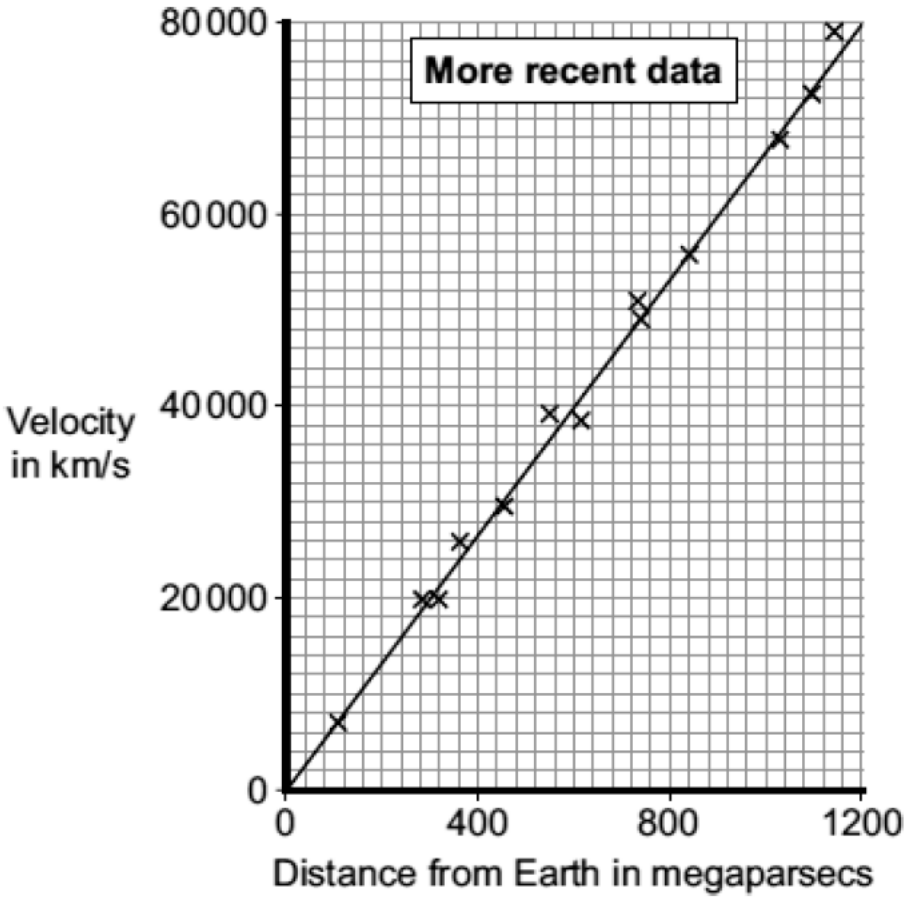
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Question 24 continues Page 16

(c) More recently, data has been obtained from more distant galaxies.



The results from the more recent data give a totally different value for the Hubble constant to the one calculated from the 1929 data.

Which set of data, the 1929 or the more recent, is most likely to give the value closest to the true value for the Hubble constant?

Draw a ring around your answer.

1929 **more recent**

Give a reason for your answer. **2**

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Question 25 (4 marks)

The ‘steady state’ theory was once a popular alternative to the ‘big bang’ theory.

The ‘steady state’ theory suggested that the universe, although expanding, had no origin and it has always existed. As the universe expands, a small amount of matter is created to keep the universe looking exactly the same all of the time.

(a) When considering the origin of the universe, what is the difference between the ‘big bang’ theory and the ‘steady state’ theory? **3**

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(b) The ‘steady state’ theory was important in encouraging new research into the universe.

Suggest a reason why scientists were keen to carry out new research. **1**

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Question 26 (3 marks)

Explain the effect of decreasing concentration on a chemical reaction using collision theory. **3**

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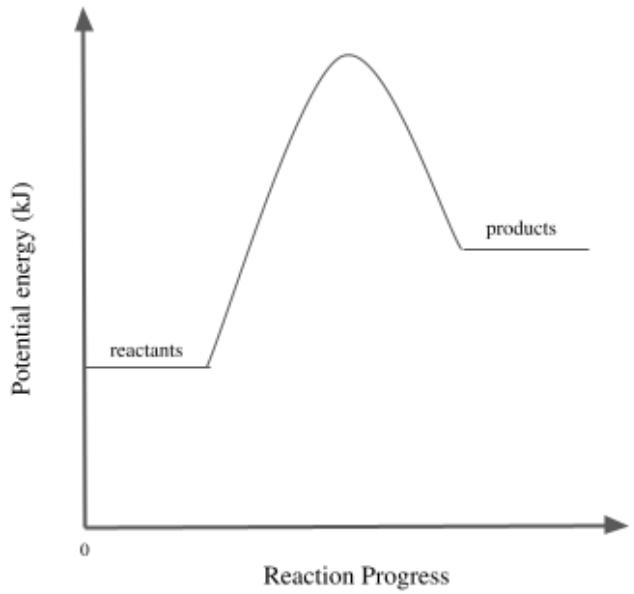
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Question 27 (3 marks)

The following diagram shows the energy profile for a reaction.



(a) Identify what type of reaction this would be in terms of energy transfer? **1**

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(b) Label on the diagram:

- The activation energy for the reaction **1**
- A possible energy profile if a catalyst was added. **1**

Question 28 (3 marks)

Outline the role of index fossils in understanding sequences of events in evolution. **3**

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Question 29 (3 marks)

Outline how a volcanic eruption can have an effect on the Earth's spheres. **3**

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Question 30 (2 marks)

Give the chemical names of TWO gases whose emissions makes major contributions to the enhanced greenhouse effect. **2**

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Question 31 (6 marks)

A student set up 2 beakers as follows:

Beaker 1 – Nickel metal in magnesium sulfate solution.

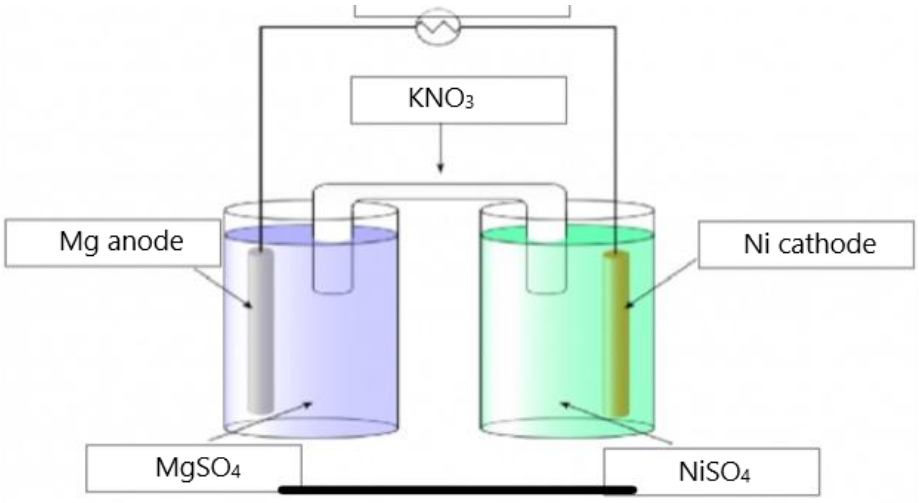
Beaker 2 – Magnesium metal in nickel sulfate solution.

They observed a metal being deposited in one of the beakers and no reaction in the other.

(a) Using the reactivity series of metals, identify which beaker had the reaction occurring. **1**

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The student then set up a galvanic cell according to the following diagram.



(b) Draw an arrow on the diagram showing the direction of electron flow. **1**

(c) Outline the purpose of the KNO_3 salt bridge. **1**

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(d) Write a balanced chemical equation for the reaction of this cell, including states. **3**

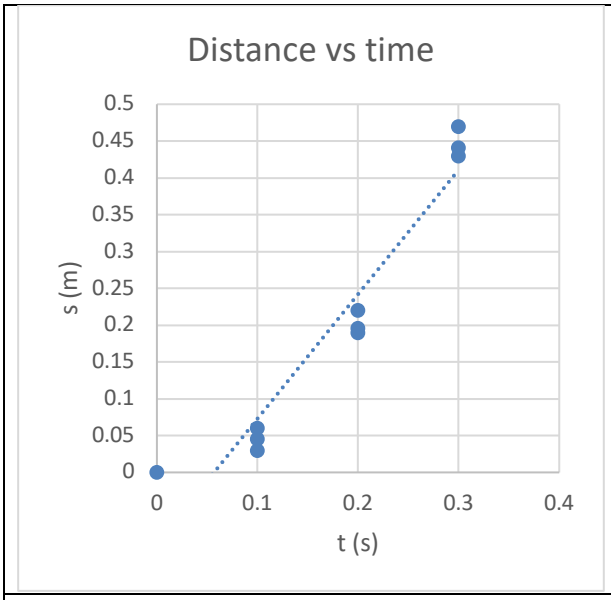
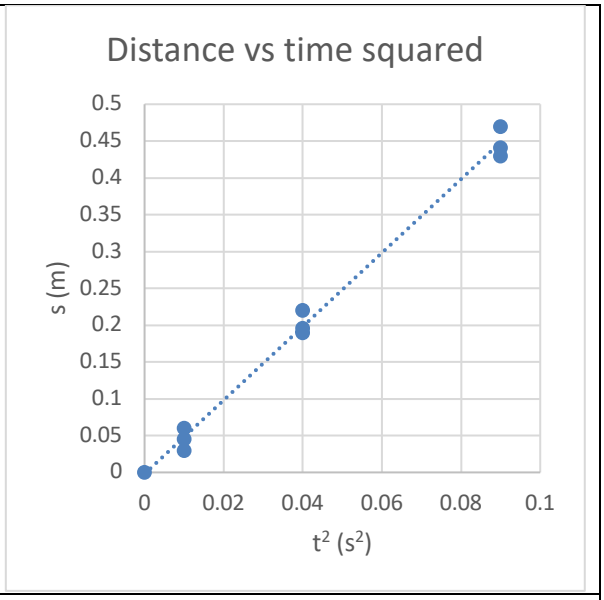
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Question 32 (10 marks)

In order to determine a relationship between variables data needs to be processed so that a straight line can be drawn through the points.

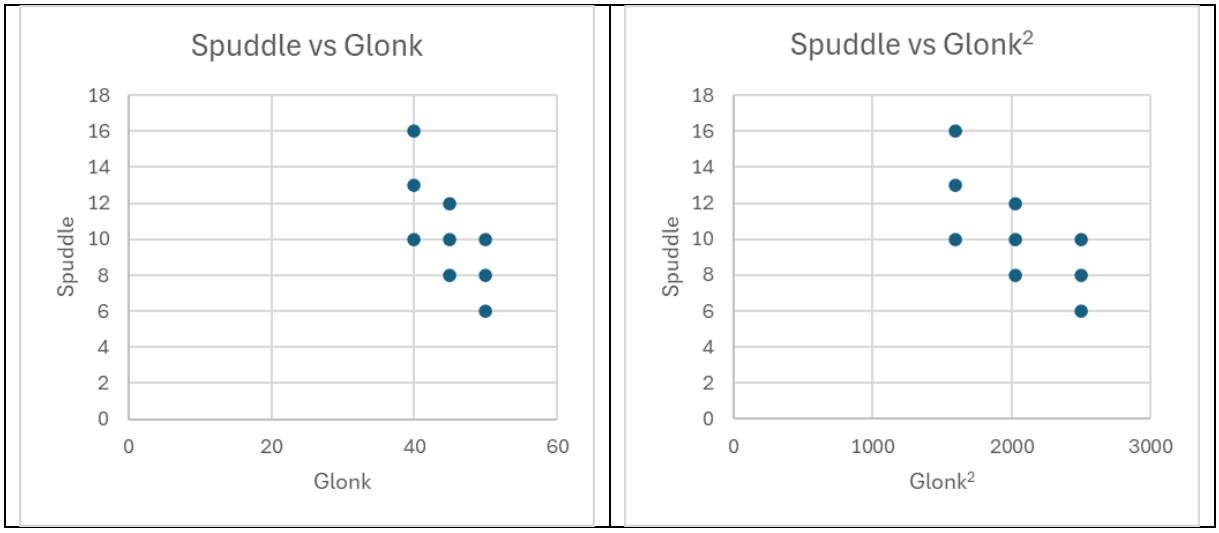
For example, to show that the distance travelled by a falling ball is proportional to time squared, the values for time would be squared, and a graph of s vs t^2 would be checked for a straight line:

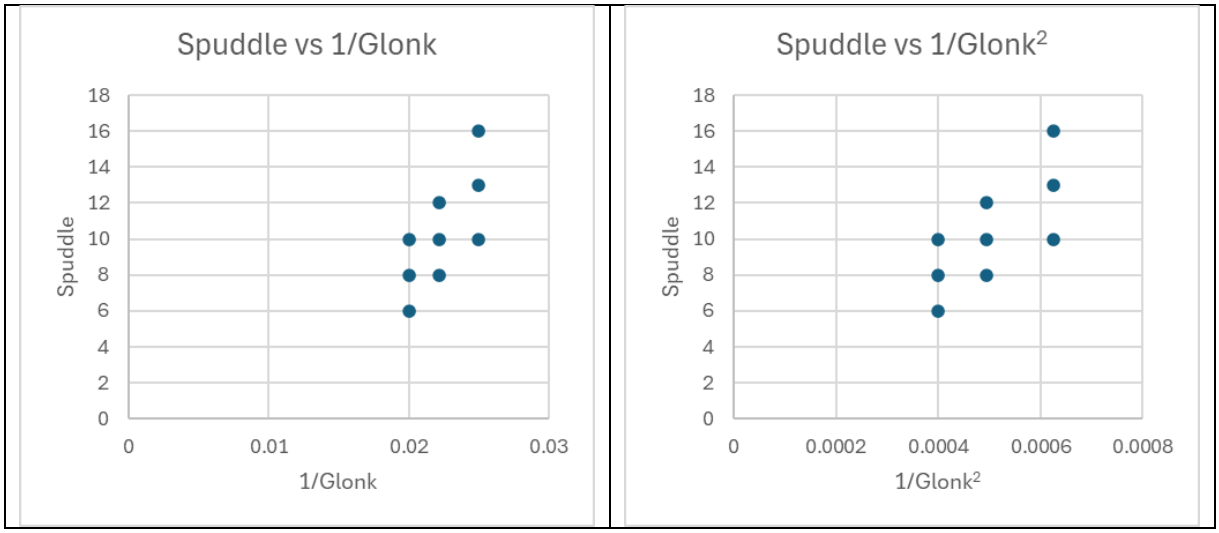
<table>
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<td>Here a straight line through the origin can not be drawn through the data. This is inconsistent with the relationship s proprortional to t. As a horizontal line cannot be drawn through the data, it is also inconsistent with no correlation between s and t.</td>
<td>Here a straight line through the origin can be drawn through the data. This is consistent with the relationship s is proportional to t^2.</td>
</tr>
</table>

The graph also shows that each data point has been repeated for a total of three times, giving a spread of data. When there is a spread of data, the line is considered to go through the data point if it passes through the range of the data points if there is less than 5 repeats. If there are 5 or more repeats, the line should go somewhere through the average ± 1 standard deviation. Given the spread of data shown above, a horizontal line cannot be drawn through the data points for the unprocessed data (s vs t graph), so there is some correlation between s and t.

Question 32 continues on page 22

Caspar from the planet Zorg is investigating the relationship between the variables glonk (G) and spuddle (S). Data is collected with three repeats, and a range of graphs plotted.





Question 32 continues on page 23

(a) What conclusion(s) could Caspar draw about the relationship between spuddle and glonk?

Annotate the graphs of spuddle and glonk to support your response. **4**

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(b) As an earthling, explain to Caspar how the data collection could be improved to better understand the relationship between spuddle and glonk.

Give THREE different suggestions. Higher marks will be awarded to better suggestions. **6**

Suggestion 1

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Suggestion 2

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Suggestion 3

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Question 33 (5 marks)

Discuss how TWO different types of evidence (other than index fossils) can be used to support our understanding of evolution. Use at least one example of each to support your response. **5**

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Question 34 (5 marks)

"Biotechnology holds transformative potential, but it is crucial that we address ethical concerns **5***proactively. The impact of our innovations on society, the environment, and individual rights must be considered as we push the boundaries of what is possible."*
– George Church

Evaluate the use of one of the following technologies and its impact on society. **5**

- Cloning
- DNA profiling
- Gene therapy
- Genetic screening
- Stem cells
- Genetic modification

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End of Examination

Extra Space Indicate (Clearly which question you are answering)

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